

E-Commerce Profitability & Marketing Efficiency Report

Analysis Period: January 2024 – December 2025

Profitability • Channel Performance • Returns • Marketing ROI

Final Draft

Executive Summary

This report analyses **2,000 orders** across eight product categories and four sales channels over the full 2024–2025 trading period. It examines category profitability, channel cost structures, return behaviour, and the return on advertising spend across six marketing platforms totalling **\$503,506** in combined spend.

2,000 Total Orders Full analysis period	\$277,969 Gross Revenue Before discounts & returns	\$236,318 Net Revenue After discounts & returns
\$56,334 Total Profit After all costs	23.84% Overall Margin Net profit / net revenue	\$20,582 Total Refunds 7.20% overall return rate
\$503,506 Marketing Spend Across 6 platforms	\$8.15M Attributed Revenue From marketing activity	16.18x Overall ROAS Revenue per \$1 spent

Key Findings

Electronics leads profitability at 31.1% margin; Books trails at 11.9% — both driven by shipping cost differences, not pricing or returns. All eight categories generate positive profit margins. Mobile App generates the highest profit per order (\$36.32) with zero platform fees. Marketplace loses 55.2% of pre-fee profit to platform charges (\$18.97/order average). Social Commerce carries the highest return rate of any channel (9.14%). TikTok Ads and Influencer Marketing deliver ROAS of 24.0x and 22.7x respectively — Email Marketing returns just \$4.81 per \$1 spent. A targeted 20% budget cut can save **\$100,569** without reducing spend on any high-performing platform.

1. Profit Margin by Product Category

Eight product categories were analysed across 2,000 orders. All categories are profitable. Profit margins range from **11.9%** (Books) to **31.1%** (Electronics), a 19-point gap driven almost entirely by shipping cost differences — not pricing, discounts, or return rates.

1.1 Margin Rankings

Category	Orders	Total Profit	Margin %	Product Cost %	Shipping %	Discount %	Return Rate
Electronics	267	\$13,973	31.13%	38.00%	13.50%	7.40%	8.61%
Toys	257	\$8,786	26.15%	39.94%	17.25%	7.72%	7.00%
Home & Kitchen	200	\$6,689	25.37%	39.99%	16.16%	7.06%	6.00%
Food & Beverage	247	\$7,592	24.76%	38.77%	17.47%	7.35%	5.67%
Sports	292	\$7,597	23.50%	41.90%	18.46%	8.32%	7.19%
Clothing	293	\$6,272	19.99%	40.90%	19.84%	7.79%	8.19%
Beauty	205	\$3,175	17.39%	41.16%	24.79%	8.39%	5.85%
Books	239	\$2,250	11.94%	39.96%	27.94%	6.51%	8.37%

1.2 What Drives the Difference?

Product costs are uniform across categories, ranging narrowly from 38.0% to 41.9% of gross revenue. Discount rates (6.5–8.4%) and return rates (5.7–8.6%) also show minimal variation. **Shipping cost as a percentage of gross revenue is the decisive variable** — ranging from 13.5% (Electronics) to 27.9% (Books).

Why Electronics Leads — 31.13% Margin

Shipping represents only 13.50% of gross revenue, the lowest of any category. High-value items absorb delivery costs efficiently. Product cost at 38.00% is the second-lowest, creating the widest gross margin in the portfolio before any other deductions.

Why Books Trails — 11.94% Margin

Shipping runs at 27.94% of gross revenue — nearly twice the Electronics rate and the highest of any category. Low average selling prices mean delivery costs dominate the cost structure. This is a structural fulfilment issue: the discount rate (6.51%) is actually the lowest in the dataset, confirming that pricing strategy is not the problem.

Why Beauty Underperforms — 17.39% Margin

A dual cost squeeze: product cost at 41.16% (above average) combined with shipping at 24.79% — the second-highest rate. Despite carrying the lowest return rate in the dataset (5.85%), high fulfilment costs chronically erode margin.

Key Findings

- Top 3 by margin: **Electronics (31.13%)**, Toys (26.15%), Home & Kitchen (25.37%).
- Bottom 3 by margin: **Books (11.94%)**, Beauty (17.39%), Clothing (19.99%).
- All eight categories generate positive profit — none are loss-making.
- Product cost (38–42%) is near-uniform; **shipping cost (13.5–27.9%) is the sole differentiator**.
- Discounts and return rates are near-identical across categories and do not explain margin gaps.

Recommendations

- Prioritise Electronics and Toys in inventory and merchandising — both exceed 25% margin with manageable return rates.
- Conduct an urgent Books shipping review. Evaluate flat-rate delivery, minimum order thresholds, or digital alternatives to reduce the 27.94% shipping burden.
- Review Beauty supplier contracts and fulfilment arrangements independently — both product cost and shipping require targeted reduction.
- Investigate which Food & Beverage SKUs drive the 9.50% revenue loss rate despite the category's low 5.67% return rate — these are disproportionately high-value returns.

2. Profitability Across Sales Channels

Four sales channels were analysed: Mobile App, Website, Marketplace, and Social Commerce. Platform fees vary significantly across channels and are the primary structural differentiator in profitability. Channels without fees retain substantially more profit per order.

2.1 Channel Performance Summary

Channel	Orders	Net Revenue	Total Profit	Profit / Order	Margin %	Avg Platform Fee	Fee Drag %
Mobile App	589	\$71,893	\$21,393	\$36.32	29.76%	\$0.00	0.0%
Website	795	\$92,991	\$25,118	\$31.60	27.01%	\$0.00	0.0%
Social Commerce	197	\$21,929	\$3,371	\$17.11	15.37%	\$9.87	36.6%
Marketplace	419	\$49,505	\$6,452	\$15.40	13.03%	\$18.97	55.2%

Fee Drag % = platform fees as a proportion of pre-fee profit. Website and Mobile App carry zero platform fees. Only Marketplace and Social Commerce incur charges.

2.2 The Platform Fee Effect

Platform fees fundamentally distort channel economics. Marketplace has strong pre-fee potential but haemorrhages profitability to platform charges.

Channel	Profit Before Fees	Avg Platform Fee	Profit After Fees	Rank (After Fees)
Mobile App	\$36.32	\$0.00	\$36.32	1st
Website	\$31.60	\$0.00	\$31.60	2nd
Marketplace	\$34.37	\$18.97	\$15.40	4th
Social Commerce	\$26.98	\$9.87	\$17.11	3rd

Best Channel: Mobile App — \$36.32 profit per order, 29.76% margin

Zero platform fees, the highest profit per order, and the second-highest order volume (589 orders). Average order value of \$140.45 with a 29.76% margin makes Mobile App the most efficient channel in the portfolio by every measure.

Worst Channel After Fees: Marketplace — \$15.40 profit per order, 13.03% margin

Without platform fees, Marketplace would generate \$34.37 profit per order — nearly matching Mobile App. Fees of \$18.97 per order on average consume 55.2% of pre-fee profit, reducing it to the worst-performing channel. Fee negotiation or a Marketplace-specific price premium should be treated as an immediate action.

Key Findings

- **Mobile App** is the best channel: \$36.32 profit/order, 29.76% margin, zero platform fees.
- **Website** is second: \$31.60 profit/order, 27.01% margin, zero fees, highest order volume (795).
- **Marketplace** is the worst after fees (\$15.40/order, 13.03%) despite pre-fee potential of \$34.37/order.
- Marketplace's fee drag of 55.2% is the largest structural cost in the channel portfolio.
- Social Commerce carries the highest return rate (9.14%) and highest revenue loss % (9.01%) of any channel.

Recommendations

- Scale Mobile App customer acquisition and UX investment — it is the highest-return channel with no fee overhead.
- Renegotiate Marketplace platform fees or apply a channel-specific price premium to offset the \$18.97/order charge.
- Redirect advertising spend toward Website and Mobile App rather than third-party Marketplace listings.
- Investigate Social Commerce return drivers — 9.14% return rate suggests lower purchase intent on this channel.

3. Return Rate and Revenue Lost

The overall return rate is **7.20%** (144 of 2,000 orders). Total refunds of **\$20,582** represent 7.40% of gross revenue. All returned orders carry net revenue of \$0 — every refund is fully absorbed by the business with no partial recovery.

3.1 Return Rate by Category

Category	Order s	Returned	Return Rate	Revenue Lost	Loss % of Gross	Risk Level
Electronics	267	23	8.61%	\$4,078	7.71%	High — rate + high item value
Books	239	20	8.37%	\$2,106	9.40%	High — highest revenue loss %
Clothing	293	24	8.19%	\$3,209	8.55%	High — volume amplifies impact
Sports	292	21	7.19%	\$2,008	5.36%	Moderate
Toys	257	18	7.00%	\$2,470	6.32%	Moderate
Home & Kitchen	200	12	6.00%	\$2,340	7.58%	Moderate — high refund values
Beauty	205	12	5.85%	\$866	4.15%	Low — lowest loss % overall
Food & Beverage	247	14	5.67%	\$3,505	9.50%	Anomalous — investigate high loss %

3.2 Return Rate by Channel

Channel	Orders	Returned	Return Rate	Revenue Lost	Loss % of Gross	Note
Social Commerce	197	18	9.14%	\$2,382	9.01%	Highest on both measures
Mobile App	589	43	7.30%	\$5,324	6.44%	Largest absolute refund total
Website	795	56	7.04%	\$9,384	8.44%	Most returned orders by volume
Marketplace	419	27	6.44%	\$3,493	6.06%	Lowest return rate

Total Revenue Lost to Returns: \$20,582 (7.40% of gross revenue)

Food & Beverage presents the most anomalous return profile: the lowest return rate (5.67%) yet the highest revenue loss percentage (9.50%), indicating that when items are returned they are disproportionately high-value. This warrants SKU-level investigation. Social Commerce is the highest-risk channel across both dimensions (9.14% rate, 9.01% revenue loss), suggesting this channel attracts lower-intent buyers. Electronics represents the single largest absolute refund loss at \$4,078, driven by the combination of the highest return rate and high average item values.

Key Findings

- Highest return rate by category: **Electronics (8.61%)**, Books (8.37%), Clothing (8.19%).
- Highest revenue loss %: **Food & Beverage (9.50%)**, Books (9.40%), Clothing (8.55%).
- By channel: Social Commerce highest (9.14%); Marketplace lowest (6.44%).
- Total of 144 returned orders resulted in \$20,582 in fully-absorbed refunds.
- A 1 percentage point reduction in return rate would save approximately \$2,800 annually.

Recommendations

- Prioritise returns reduction for Electronics — improve product specifications, technical details, and compatibility guides.
- Introduce size guides and fit information for Clothing across all channels to reduce the 8.19% return rate.
- Conduct SKU-level analysis on Food & Beverage returns — identify which items generate the disproportionate 9.50% revenue loss.
- Analyse Social Commerce buyer demographics and behaviour — the 9.14% return rate is 42% higher than Marketplace.
- Consider a returns monitoring dashboard tracking rate, revenue lost, and avg refund value monthly by category and channel.

4. Marketing Spend & ROAS by Platform

Six advertising platforms were active across the 24-month analysis period. Total spend: **\$503,506**. Total attributed revenue: **\$8.15M**. Overall ROAS: **16.18x**. Performance varies from **4.81x (Email Marketing)** to **24.02x (TikTok Ads)** — a five-fold gap between the best and worst platforms.

4.1 ROAS Rankings by Platform

Platform	Total Spend	Revenue Attributed	ROAS	Avg CPA	Spend %	Revenue %	Efficiency Gap
TikTok Ads	\$57,229	\$1,374,627	24.02x	\$3.93	11.4%	16.9%	+5.5%
Influencer	\$97,663	\$2,216,974	22.70x	\$4.80	19.4%	27.2%	+7.8%
Instagram Ads	\$65,154	\$1,024,639	15.73x	\$5.55	12.9%	12.6%	-0.4%
Google Ads	\$152,546	\$2,194,121	14.38x	\$6.48	30.3%	26.9%	-3.4%
Facebook Ads	\$106,452	\$1,218,572	11.45x	\$8.38	21.1%	15.0%	-6.2%
Email Marketing	\$24,461	\$117,681	4.81x	\$26.01	4.9%	1.4%	-3.4%

Efficiency Gap = Revenue Share % minus Spend Share %. Positive: platform earns more revenue than its proportional share of budget. Negative: it costs more than it returns proportionally.

4.2 Efficiency Classification

Classification	Threshold	Platforms	Action
Efficient	ROAS ≥ 20x	TikTok Ads (24.02x), Influencer (22.70x)	Grow — protect and increase budget
Acceptable	ROAS 15–20x	Instagram Ads (15.73x)	Hold — maintain current spend level
Below Target	ROAS 10–15x	Google Ads (14.38x), Facebook Ads (11.45x)	Reduce — cut 15–50% based on gap
Underperforming	ROAS < 10x	Email Marketing (4.81x)	Cut — eliminate and reallocate

Standout Performers: TikTok Ads (24.02x ROAS) and Influencer (22.70x ROAS)

These two platforms account for 30.8% of total spend but generate 44.1% of attributed revenue. TikTok's CPA of \$3.93 is the lowest of any platform — it acquires customers more cheaply than any other channel. Influencer has the best efficiency gap at +7.8%: every dollar invested generates disproportionately high revenue relative to its budget share.

Underperformers: Facebook Ads (-6.2% gap) and Email Marketing (4.81x ROAS)

Facebook consumes 21.1% of total budget but generates only 15.0% of revenue — the worst efficiency gap in the portfolio. Email Marketing has a ROAS of just 4.81x and a CPA of \$26.01, which is 6.6x TikTok's acquisition cost. Google Ads holds 30.3% of budget but generates only 26.9% of revenue, indicating its spend has grown beyond the point of proportional return.

Key Findings

- **TikTok Ads:** best ROAS (24.02x), lowest CPA (\$3.93), +5.5% efficiency gap.
- **Influencer:** second-best ROAS (22.70x), best efficiency gap (+7.8%) — 19.4% of spend generates 27.2% of revenue.
- **Email Marketing:** worst performer at 4.81x ROAS and \$26.01 CPA — 6.6x TikTok's acquisition cost.
- **Facebook Ads:** worst efficiency gap (-6.2%) — 21.1% of budget generates only 15.0% of attributed revenue.
- Google Ads holds the single largest budget share (30.3%) but under-delivers by 3.4 percentage points on revenue.

Recommendations

- Protect TikTok Ads and Influencer budgets unconditionally — any future budget pressure must come from underperforming platforms.
- Reduce Facebook Ads spend significantly — the -6.2% efficiency gap is the largest drag in the entire portfolio.
- Eliminate Email Marketing budget entirely and redirect to TikTok Ads or Influencer where returns are proven.
- Trim Google Ads to right-size its spend allocation to match its revenue contribution.
- Set a ROAS floor of 10x as the minimum acceptable performance threshold for continued investment.

5. 20% Budget Cut Recommendation

Total marketing budget: **\$503,506**. A 20% reduction target requires saving **\$100,701**. The recommended strategy concentrates all cuts on the three lowest-performing platforms, leaving TikTok Ads, Influencer Marketing, and Instagram Ads fully funded and protected.

5.1 Recommended Platform Cut Plan

Platform	ROAS	Current Spend	Cut %	Saving	New Budget	Rationale
Email Marketing	4.81x	\$24,461	100%	\$24,461	\$0	ROAS below 5x — eliminate and reallocate entirely
Facebook Ads	11.45x	\$106,452	50%	\$53,226	\$53,226	Worst efficiency gap (-6.2%) in the portfolio
Google Ads	14.38x	\$152,546	15%	\$22,882	\$129,664	Right-size spend to match revenue contribution
Instagram Ads	15.73x	\$65,154	0%	\$0	\$65,154	Acceptable ROAS — hold steady
Influencer	22.70x	\$97,663	0%	\$0	\$97,663	Strong ROAS +7.8% gap — protect
TikTok Ads	24.02x	\$57,229	0%	\$0	\$57,229	Highest ROAS, lowest CPA — do not cut
TOTAL	16.18x	\$503,506	20%	\$100,569	\$402,936	Meets \$100,701 target — within 0.1%

5.2 Priority Month × Platform Cuts

Within the platforms being reduced, these specific month/platform combinations represent the highest spend at the lowest ROAS. These are the first line items to pause or reduce when implementing the cut plan.

Platform	Month	Spend	ROAS	Priority	Recommended Action
Email Marketing	Dec 2025	\$1,797	0.67x	1 — Cut	Suspend immediately — returns \$0.67 per \$1 spent
Email Marketing	Oct 2025	\$792	1.21x	2 — Cut	Near-zero return — suspend
Email Marketing	Jul 2025	\$1,362	1.74x	3 — Cut	Below 2x ROAS — suspend
Facebook Ads	Nov 2024	\$8,556	6.60x	4 — Cut	Largest single Facebook loss month

Platform	Month	Spend	ROAS	Priority	Recommended Action
Facebook Ads	Nov 2025	\$6,884	4.44x	5 — Cut	Repeat poor November — avoid this month seasonally
Facebook Ads	Sep 2025	\$3,878	2.51x	6 — Cut	Very low ROAS — reduce aggressively
Google Ads	Sep 2025	\$6,501	3.84x	7 — Trim	Worst Google month — large spend, lowest ROAS
Google Ads	Jul 2025	\$5,844	4.27x	8 — Trim	Second worst Google month — high spend, low return

Budget Cut Validation — Target Met

Email Marketing (100% cut): \$24,461 saved Facebook Ads (50% cut): \$53,226 saved Google Ads (15% cut): \$22,882 saved Total savings: \$100,569 — meets the \$100,701 target within 0.1% TikTok Ads, Influencer, and Instagram Ads are left entirely untouched. These three platforms generate 56.7% of all attributed revenue from 44.6% of total spend.

Key Findings

- The \$100,701 target is achievable by cutting only three platforms — no high-ROAS channel needs to be touched.
- Email Marketing should be eliminated entirely (ROAS 4.81x, CPA \$26.01 — unacceptably poor by any benchmark).
- November is consistently the worst month for Facebook Ads in both 2024 and 2025 — a seasonal pattern requiring pre-planned cuts.
- September and July are consistently the worst months for Google Ads — both years show ROAS below 5x at high spend levels.

Recommendations

- Implement cuts in order of priority: Email Marketing first (immediate), Facebook November months second, Google September/July third.
- Reallocate the \$24,461 from Email Marketing to TikTok Ads or Influencer where return on investment is proven.
- Build seasonal spend floors into the annual media plan: Facebook budgets should automatically reduce in November; Google in September and July.
- Review the remaining Facebook and Google spend quarterly against the 10x ROAS floor to prevent further budget creep.
- Reassess Email Marketing channel entirely — consider whether the customer segment it reaches can be reached more efficiently through Influencer or TikTok.

6. Strategic Recommendations Summary

The following table consolidates the top recommendations across all five areas of analysis.

Area	Recommendation	Expected Impact	Priority
Product	Review Books shipping — flat-rate or threshold delivery	Margin uplift from 11.9% toward 18–20%	High
Product	Prioritise Electronics + Toys in inventory allocation	Protect highest-margin categories	High
Product	Review Beauty supplier + fulfilment contracts	Reduce 24.79% shipping cost burden	Medium
Product	SKU-level investigation: Food & Bev high-value returns	Reduce 9.50% revenue loss rate	Medium
Channels	Scale Mobile App acquisition — zero fees, 29.76% margin	Increase highest-profit-per-order volume	High
Channels	Renegotiate Marketplace fees or add price premium	Recover portion of 55.2% fee drag	High
Channels	Analyse Social Commerce return behaviour	Reduce 9.14% return rate	Medium
Returns	Add size guides / product specs for Clothing + Electronics	Target 8%+ return rates in both categories	Medium
Marketing	Eliminate Email Marketing (\$24,461 saved)	Redirect to TikTok/Influencer at 5–6x ROAS	Immediate
Marketing	Cut Facebook Ads 50% (\$53,226 saved) — Nov months first	Remove worst efficiency gap (-6.2%)	Immediate
Marketing	Trim Google Ads 15% (\$22,882 saved) — Sep/Jul months first	Right-size to revenue contribution	Immediate
Marketing	Protect TikTok (24.02x) and Influencer (22.70x) budgets	Preserve 44.1% of attributed revenue	Ongoing

Appendix: Data Quality Notes

The following data quality observations were confirmed during the validation phase prior to analysis.

- **Cost reconciliation:** Zero mismatches across all 2,000 orders. $\text{product_cost} + \text{shipping_cost} + \text{platform_fee} + \text{transaction_fee} = \text{total_costs}$ confirmed for every row.
- **Negative profit orders:** 563 orders (28.2%) record negative profit. This reflects loss-making transactions where fulfilment costs exceed net revenue and is confirmed as expected business behaviour, not a data error.
- **Returned field:** Stores string values ('Yes'/'No') rather than booleans. All return rate SQL queries use `CASE WHEN returned = 'Yes' THEN 1 ELSE 0 END`.
- **Returned orders revenue:** All 144 returned orders record $\text{net_revenue} = \$0$ exactly, confirming full refunds are applied. Revenue loss calculations use gross_revenue as the denominator throughout to avoid division-by-zero.
- **Platform fees:** Website and Mobile App record zero platform fees across all 1,384 combined orders. Only Marketplace and Social Commerce (616 orders combined) incur platform charges.
- **marketing_spend schema:** The month column stores YYYY-MM format strings. The column type must be altered to `VARCHAR(10)` before a composite primary key can be applied in MySQL.
- **Products table:** The products table (207 rows) was not directly joined in order-level analysis. The `primary_category` field in the orders table provides sufficient category granularity for all five business questions.
- **Duplicate key check:** No duplicate `order_id` values found in orders. No duplicate `product_id` values found in products. No duplicate month/platform combinations found in `marketing_spend`.